

Question Paper

Exam Date & Time: 29-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026
SUBJECT: BIO 1071/BIO_1071 BIOLOGY FOR ENGINEERS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Explain why silicon is considered a potential alternative to carbon as a basis for alien life, despite its limitations on Earth. Provide your answer Only in the following format. (3)

Sl No.	Criterion	Explanation
1		
2		
3		

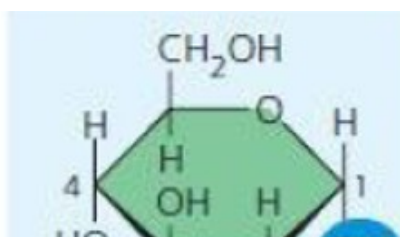
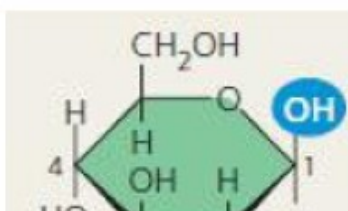
Course Outcome: "CO1" / Cognitive Level: "Understand"

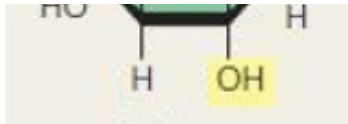
- 1B) When a protein is treated with a strong acid, which structural level is most likely to be affected initially? Justify Provide your answer in the following format: (3)

Sl No	Structural Level Affected	Affected or not with justification
1	Primary	
2	Secondary	
3	Tertiary	
4	Quaternary	

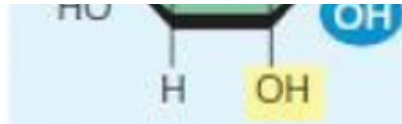
Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) (4)





A



B

i) Illustrate how the above polymers differ in their bonding?

ii) Analyze the differences of the properties of the polymer because of the specific bonding

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2A) Draw and interpret a Punnett square for a dihybrid test cross involving seed shape (Rr) and pod color (Gg), where round seed shape (R) and green pod color (G) are dominant traits. Cross this with a plant of genotype rrgg. Explain the possible genotypes and phenotypes of the offspring, highlighting which combinations express the dominant traits. (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2B) (a) Determine the recombination frequency between factors for leaf shape and stem color in a plant species. (3)
Dominant traits: broad leaves and green stems.
Given offspring from a test cross:

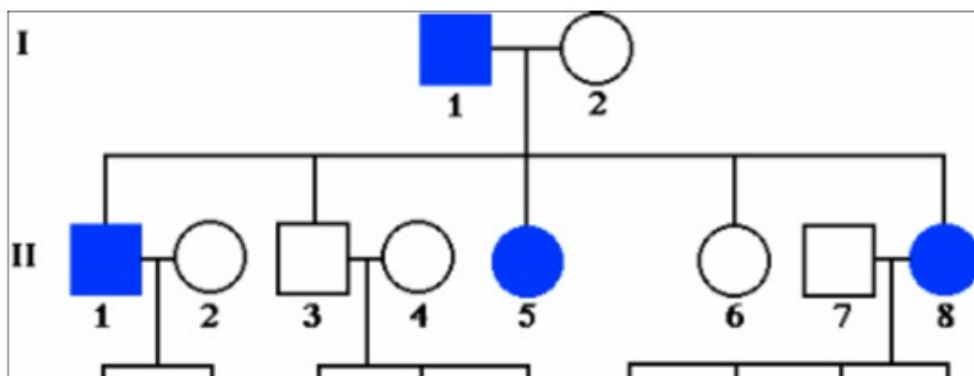
- 920 broad-leaved green-stemmed
- 910 narrow-leaved purple-stemmed
- 180 broad-leaved purple-stemmed
- 190 narrow-leaved green-stemmed

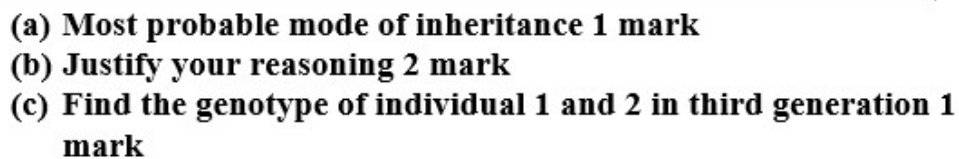
(b) Explain what this frequency indicates about the genetic distance between these traits.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) (4)

Evaluate the pedigree chart and answer the questions





3A) Calculate the proportion of heavy, hybrid, and light DNA strands after four generation of replication in an alien bacteria originally grown in ^{15}N and then transferred to ^{14}N medium. Show your calculation. Mode of replication = Semi conservative (3)

3B) Defend the conclusion that DNA is the transforming principle. If Avery were on a reality show called "*Molecules Got Talent*", would DNA win the trophy? Why? Discuss your answer based on talent, versatility and impact. (3)

3C) (4)

The diagram illustrates the central dogma of molecular biology, showing the flow of genetic information from DNA to RNA to Protein.

DNA: A double helix structure is shown with the sequence: ATGCTTTCGTAT (top strand) and TACGAAAGCA (bottom strand). The sequence is repeated three times.

Transcription: An arrow points from the DNA to the RNA, indicating the process of transcription.

RNA: A single-stranded molecule is shown with the sequence: AUGCUUUCGUU (top strand) and UACGAAAGCAU (bottom strand). The sequence is repeated three times.

Translation: An arrow points from the RNA to the Protein, indicating the process of translation.

Protein: A chain of amino acids is shown, represented by orange circles. The sequence is: AUGCUUUCGUU (top strand) and UACGAAAGCAU (bottom strand). The sequence is repeated three times, followed by a red "STOP" codon.

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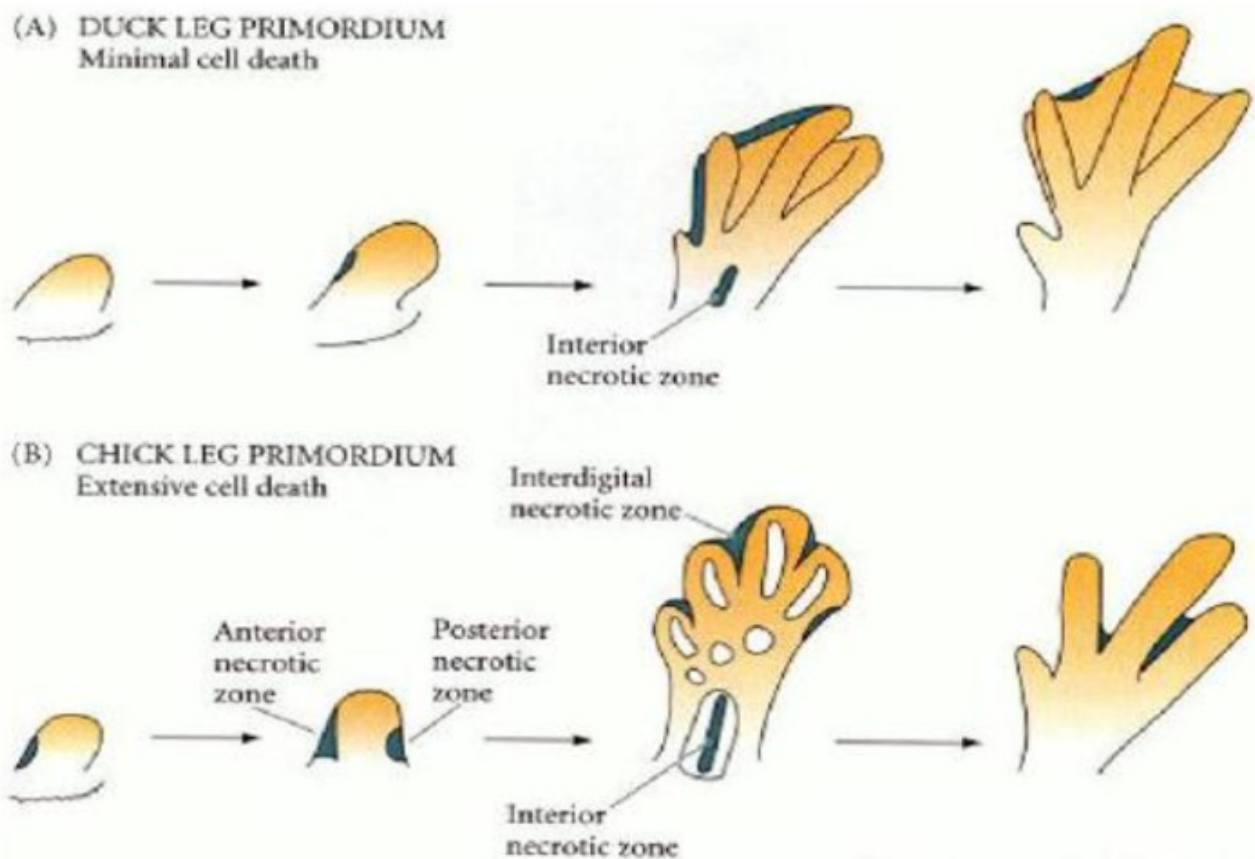
- 4A) Explain the concepts; (3)
 i) Hermit crab and Sea anemone.
 ii) +/- interaction with example
 (1.5+1.5 = 3 marks)

Course Outcome: "CO4" / Cognitive Level: "Understand"

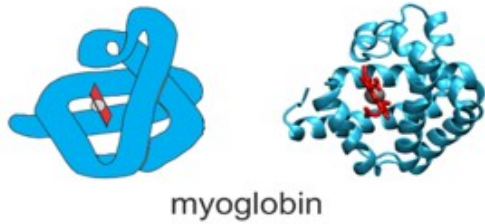
- 4B) According to the theory, proposed by French naturalist Jean-Baptiste Lamarck in the early 19th century, suggested that (3)
 organisms could acquire new characteristics during their lifetime in response to their environment and needs, and then
 pass these acquired traits on to their offspring. Explain with examples.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4C) i) How does the embryonic development of webbed feet provide evidence for evolution? (4)



ii) How can the structure of Myoglobin be considered as genetic evidence?



(2+2 = 4 marks)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5A) (I) Why do we need Booster dose after vaccination? (3)
(II) Which type of vaccine is prepared by typically killing the pathogen by using heat or chemicals like formalin?
(III) When the vaccine virus is given to a human, it will be unable to replicate enough to cause illness but will still provoke an immune response that can protect against future infection. Name the type of vaccine.
(1+1+1 = 3 marks)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5B) Researchers have discovered a technique to extract drinkable water, inspired by namib beetle. What is the concept behind this? (3)



Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5C) What is the concept for bioinspiration in the following cases? Explain. (4)

Sl N o	Concept	Explanation
1		
2		
3		
4		

-----End-----

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	BIO_1071:BIOLOGY FOR ENGINEERS

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	BIO_1071:BIOLOGY FOR ENGINEERS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	6	20(40%)
3	Application	3	9(18%)
4	Analysis	4	13(26%)
5	Create	2	8(16%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)